

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2013-14

Zoology

Paper: ZOB-602 — Cell Biology, Genetics and Evolution

Time: Three hours

Full Marks: 70

Note: Answer two questions from each Section including Question No. 1 of each Section which is compulsory. Use a separate answer book for each Section. The figures in the right-hand margin indicate marks.

SECTION – A
(Cell Biology) (Marks: 26)

Note: Answer two questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

(a) Write whether the following statements are True or False, giving reasons for your answer in 3–4 lines: [2 × 2 = 4]

(i) The endoplasmic reticulum lumen is full of soluble Luminal resident proteins and enzymes, whereas the resident proteins in the Golgi apparatus are all membrane bound.

(ii) Lysosome contains only hydrolases.

(b) Select the most suitable answer for the following: [2 × $\frac{1}{2}$ = 1]

(i) Multivesicular Bodies (MVBs) are found in:

(1) early endosomes

(2) late endosomes

(3) hybrid endosomes

(4) lysosomes

(ii) KDEL protein is retained in:

(1) cytoplasm

(2) nucleus

(3) endoplasmic reticulum

(4) Golgi apparatus

(c) Differentiate between the following: [2 × $1\frac{1}{2}$ = 3]

(i) Retrograde transport and Anterograde transport

(ii) Active transport and Passive transport.

Question 2. Answer the following parts:

- (a) Define signal peptides. Explain entry and passage of proteins through ER. $[2 + 7 = 9]$
- (b) Explain the events in different phases of cell cycle. Describe the genetic regulation of the cell cycle. $[3 + 6 = 9]$

Question 3. Answer the following parts:

- (a) Describe the structure of Lampbrush chromosome and explain its significance in the study of gene expression. $[4 + 5 = 9]$
- (b) Write short notes on the following: $[3 \times 3 = 9]$
 - (i) Centromere
 - (ii) Nuclear pore complex
 - (iii) Endosomes.

SECTION – B
(Genetics) (Marks: 26)

Note: Answer two questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

- (a) Write whether the following statements are True or False giving valid reasons for your answer: [2 × 2 = 4]
- (i) The on/off switch of *sex* gene of *Drosophila* is not sensitive to the X : A ratio.
 - (ii) Frameshift mutations are more detrimental than base substitution mutations.
- (b) The promoter region of an operon has a mutation to which RNA polymerase cannot bind and initiate transcription. Will this mutation prevent the binding of RNA polymerase to another copy of operon in trans condition? Explain with reasons. [2]
- (c) Select the best options from the choices given below: [4 × $\frac{1}{2}$ = 2]
- (i) Out of 800 progeny of a three-point testcross there were 12 double crossover recombinants, whereas 20 had been expected on the basis of no interference. The interference is:
 - (1) 0.10
 - (2) 0.20
 - (3) 0.30
 - (4) 0.40
 - (ii) What will be the state of lac operon of a *lac I* mutant?
 - (1) The *lac* genes would be expressed efficiently only in the absence of lactose.
 - (2) The *lac* genes would be expressed efficiently only in the presence of lactose.
 - (3) The *lac* genes would be expressed continuously.
 - (4) The *lac* genes would never be expressed efficiently.
 - (iii) Phenylketonuria (PKU) arises because of:
 - (1) inclusion bodies in the neurons
 - (2) metabolic dysfunction
 - (3) multifactorial causes
 - (4) transposable element
 - (iv) A strain of *E. coli* is Lac[−] and Met[−]. After transformation experiment to select only for Met⁺ cells, it should be plated on:
 - (1) minimal media + glucose + methionine
 - (2) minimal media + glucose
 - (3) rich media

- (4) minimal media + lactose + methionine

Question 2. Answer the following parts:

- (a) Amino acid glycine is encoded by CGA. A single base substitution changes it to a nonsense codon. Specify the base change that would have taken place to convert it into a nonsense codon. Is this change a transversion or transition? Distinguish between the two. [1 + 1 + 2 = 4]
- (b) Two-point testcrosses revealed the following map results:
- (i) A — B 17 map units
 - (ii) B — C 11 map units.

What are the two possible maps involving the three given loci? Which additional cross should be performed to resolve the correct map? [1 + 3 = 4]

- (c) An F' (F prime) element containing the maltose gene (*mal*) is found in *E. coli*. Explain in stepwise manner how this element would have been produced starting from an F⁺ plasmid. [6]
- (d) What are DNA arrays? How are they useful in studying gene expression? [4]

Question 3. Answer the following parts:

- (a) The height of a plant is controlled by two pairs of alleles assorting independently, A, a and B, b. Each dominant allele, A and B contribute 40 cm to the height and each recessive alleles a and b contribute 10 cm to the height. Assuming that these loci have additive effects, what will be the height of the F₁ progeny from a cross of AABB × aabb? How many phenotypic classes are possible in F₂ progeny? And also mention the range of height of F₂ progeny. [2 + 3 = 5]
- (b) Describe the three types of tetrad that can be obtained from tetrad analysis of *Neurospora*. How can one distinguish whether the two genes are linked or on different chromosomes on the basis of type of the tetrads observed? [2 + 3 = 5]
- (c) A transposable element has a deletion in the transposase gene. What will be the consequence of this deletion? Under what conditions can this transposable element jump in the genome? [3]
- (d) Briefly write on triplet repeat expansion disorder taking Huntington's disease as an example. How is it different from a multifactorial disease? [3 + 2 = 5]

SECTION – C
(Evolution) (Marks: 18)

Note: Answer two questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False, giving reasons in 2–3 sentences:

[2 × 2 = 4]

(i) Toothed birds originated in Cambrian period and became extinct by the end of Cretaceous period.

(ii) Eohippus possessed 3 toes in fore- and hind-limbs.

- (b) Define the following:

[2 × 1 = 2]

(i) Mullerian mimicry

(ii) Hardy Weinberg law of equilibrium.

Question 2. Answer the following parts:

- (a) What is molecular phylogeny? How can DNA sequences or proteins be used to establish phylogenetic relationships among the different group of animals? [6]

- (b) Explain the following:

[2 × 3 = 6]

(i) Radioactive dating

(ii) Adoptive colouration.

Question 3. Answer the following parts:

- (a) Describe the various means of prezygotic reproductive isolation.

[6]

- (b) Write short notes on the following:

[2 × 3 = 6]

(i) Allopatric speciation

(ii) Chromosomal polymorphism.

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